***Brock Biology of Microorganisms, 15e* (Madigan et al.)**
Chapter 23 Microbial Symbioses

23.1 Multiple Choice Questions

1) Two organisms that both benefit from each other are in a symbiotic relationship called
A) ammensalism.
B) commensalism.
C) mutualism.
D) parasitism.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.1

2) The insect symbiont *Wolbachia* can be used to suppress disease transmission by mosquitoes because
A) *Wolbachia* supplies essential nutrients to the mosquito *Culex quinquefasciatus.*
B) only *Wolbachia*-infected mosquitoes can transmit viral diseases such as dengue fever.
C) *Wolbachia* infections can spread rapidly through a mosquito population, killing all the mosquitoes.
D) *Wolbachia*-infected male mosquitoes sterilize uninfected female mosquitoes.
Answer: D
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 23.6

3) Most nitrogen-fixing bacteria symbiotically associated with plants are called
A) bacteroids.
B) mycorrhizae.
C) rhizobia.
D) symbiodinia.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.3

4) Plant root nodules are
A) formed from gall-rotting bacteria that decompose plant roots.
B) harmful to plants, because the bacteria outcompete the plants for nutrients.
C) sites where nitrogen fixation occurs.
D) structures created by fungi and are found in all agricultural crops.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.3

5) The function of leghemoglobin in root nodules is to
A) bind oxygen.
B) chelate iron.
C) produce iron.
D) produce nitrogen.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.3

6) What is the primary difference between a bacteroid a bacterial cell?
A) There are genetic differences that can be detected using 16S rRNA gene sequencing.
B) They are biochemically very different from bacterial cells.
C) Their cell morphology is different from bacterial morphology.
D) They have genetic differences that can be detected using DNA sequencing.
Answer: C
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 23.3

7) The specificity of certain rhizobia to infect only particular plants is in part due to the
A) abundance of nutrients present in the soil.
B) *nifH* genes they possess.
C) rhizobial lipids that act as signaling molecules.
D) season (time of year).
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.3

8) The mutualism of *Azolla-Anabaena* is useful for
A) aquaculturalists.
B) corn farmers.
C) rice farmers.
D) tropical forest community succession.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.3

9) Which bacterium contains a large Ti plasmid and causes crown gall disease in plants?
A) *Agrobacterium rhizogenes*
B) *Agrobacterium tumefaciens*
C) *Aliivibrio fischeri*
D) *Butyrivibrio fibrisolvens*
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.5

10) What is the role of the *vir* genes on a Ti plasmid?
A) to cause crow gall disease (virulence)
B) to confer resistance to viral infection
C) to initiate replication of the plasmid
D) to allow T-DNA transfer
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.5

11) Where are ectomycorrhizae most highly developed?
A) boreal and temperate forests
B) tundra
C) tropical dry forests
D) tropical rain forests
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.4

12) How do arbuscular mycorrhizae (AM) help plants obtain more nutrients from the soil?
A) By-products of AM metabolism serve as primary nutrient sources for the plants.
B) The AM increase the total surface area to absorb more nutrients.
C) The hyphae develop as specialized nodules that directly produce nutrients for the plant.
D) Signaling molecules in the plant are passed to the AM to initiate electron flow, which is in turn used to create ATP.
Answer: B
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 23.4

13) The rumen is an ________ habitat that depends on ________ to digest cellulose for ruminant animals.
A) aerobic / cellulolytic fungi
B) aerobic / cellulolytic and fermentative bacteria
C) anaerobic / methanogens
D) anaerobic / cellulolytic and fermentative bacteria
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.13

14) Which compounds serve as major sources of energy for ruminant animals?
A) acetic acid and glucose
B) acetic, butyric, and propionic acids
C) dimethylsulfoniopropionate, gluconate, and protocatechuate
D) glucose and sucrose
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.13

15) Myc factors are
A) a type of strigolactone hormone.
B) lipochitin oligosaccharide signaling proteins produced by arbuscular mycorrhizae.
C) examples of chemicals that mycorrhizae obtain from the soil.
D) Carbohydrate structures that form connections between mycorrhizae and plants.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.4

16) The fungal division containing arbuscular mycorrhizae is the
A) *Betaproteobacteria*.
B) *Basidiomycota*.
C) *Glomerulomycota*.
D) *Rhizobia*.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.4

17) *Frankia* are
A) a type of arbuscular mycorrhizae.
B) glomerulomycetes that live in root nodules.
C) a species of tree that has root nodules.
D) actinomycetes that fix nitrogen.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.3

18) In insects, primary symbionts are ________, while secondary symbionts are ________.
A) heritable / not heritable
B) able to replicate outside of the host / obligate symbionts
C) required for host reproduction / not required for host reproduction
D) obligate symbionts / sometimes free-living
Answer: C
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 23.6

19) Horizontal gene transfer between insects and their symbionts
A) happens frequently because of coevolution.
B) causes genome reduction in the symbiont and genome expansion in the host.
C) causes genome reduction in both the host and the symbiont.
D) has been observed, but is considered a rare event.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.6

20) Which of the following is a common feature of primary insect symbionts?
A) accelerated mutation rates only
B) extreme genome reduction only
C) high A+T content only
D) accelerated mutation rates, extreme genome reduction, and high A+T content
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.6

21) Bacterial symbionts of termite hindguts play a critical role in the termite′s
A) reproduction.
B) nitrogen metabolism.
C) resistance to fungal infection.
D) reproduction and nitrogen metabolism.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.7

22) Symbionts such as *Rickettsia* in whiteflies and *Buchnera* in aphids are transmitted ________ to ensure the safe transfer of the symbiont to the next host generation.
A) vertically to offspring
B) horizontally through infected water
C) horizontally through infected secondary hosts
D) vertically through the air
Answer: A
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 23.6

23) Which of the following do dinoflagellates most commonly associate with in a mutualistic relationship?
A) clams
B) corals
C) flatworms
D) snails
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.12

24) What is the primary benefit to phototrophic microorganisms associated with corals?
A) Calcium and carbonate ions are released, which help buffer the pH and prevent dramatic pH shifts.
B) The coral skeleton is used as a source of bicarbonate for the autotrophic growth of the phototrophs.
C) The coral skeleton is an extremely efficient light-gathering structure that greatly enhances light harvesting for photosynthesis.
D) The water temperature in and around the coral skeleton is much warmer than elsewhere.
Answer: C
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 23.12

25) Ruminants digest bacterial cells as a primary source of
A) carbohydrates.
B) vitamins and proteins.
C) proteins.
D) vitamins.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.13

26) Two main symbionts of termites are ________ and ________.
A) archaea / bacteria
B) bacteria / fungi
C) bacteria / protists
D) fungi / protists
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.7

27) What percentage of terrestrial plants are colonized by arbuscular mycorrhizae?
A) 10-15%
B) 25-30%
C) 50-60%
D) 70-90%
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.4

28) What is one reason that acetogenesis is favored over methanogenesis in the termite gut?
A) The environment is anoxic.
B) Methanogens can use a wide range of substrates, but acetogens specialize in those produced in the termite gut.
C) Methanogens are not found in the termite gut.
D) Acetogens are better at colonizing the center of the gut, which is rich in H_2.
Answer: D
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 23.7

29) Which of the following are important adaptations specifically to herbivores?
A) having a large, anoxic chamber where fermentation can occur and having an extended retention time for food
B) having a large, oxic chamber where fermentation can occur and having an extended retention time for food
C) having a large, anoxic chamber where fermentation can occur and having relatively rapid food passage through the digestive tract
D) having a large, oxic chamber where fermentation can occur and having relatively rapid food passage through the digestive tract
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.12

30) Why do obligate symbionts often contain lower G+C content when compared to free-living organisms?
A) Ancestral symbionts had low GC content genomes by chance, and these low GC content genomes are passed onto their progeny.
B) Organisms with a low G+C content organisms are at a selective disadvantage to those with a high G+C content when free-living.
C) It is more difficult to replicate high G+C content genomes than to replicate low G+C content genomes.
D) Two common spontaneous mutations change GC pairs into AT pairs and symbionts usually have fewer DNA repair mechanisms to fix these compared with free-living organisms.
Answer: D
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 23.6

31) A species of insect that lives solely on pine sap was found to contain a bacterial symbiont. Genome sequencing of both the host and the symbiont revealed that the symbiont lacked many genes required for energy production, and the host lacked genes for biosynthesis of several essential amino acids. What is/are the most likely mechanism(s) that caused the loss of these genes?
A) coevolution
B) horizontal gene transfer
C) increased mutation rate
D) horizontal gene transfer and increased mutation rate
Answer: A
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 23.6

32) When squids that contain luminescent bacterial symbionts hatch, they do NOT contain a bacterial symbiont. Symbiont transmission in this symbiosis is
A) vertical from parent to offspring.
B) horizontal and involves specific selection of the symbiont from the environment.
C) random and results in various different species being selected as the symbiont.
D) mixed and results in multiple symbiont species colonizing the squid at once.
Answer: B
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 23.8

33) Which of the following is a common benefit of a microbe-plant symbiosis?
A) increased nutrient availability only
B) decreased pathogen colonization only
C) increased affinity for carbon dioxide
D) increased nutrient availability and decreased pathogen colonization
Answer: D
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 23.3

34) How are root nodule and arbuscular mycorrhizal symbioses similar?
A) They both use lipochitin oligosaccharide signaling factors to initiate root colonization.
B) They both supply nitrogen to the host plant through nitrogen fixation.
C) They both increase absorption of nutrients from soil.
D) They are both required for the growth and reproduction of their host plants.
Answer: A
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 23.4

35) A mutant of *Rhizobium leguminosarum* is able to survive and reproduce in the laboratory outside of plant roots, but can no longer initiate root nodule formation. What type of genes are most likely mutated in this mutant?
A) *rhz* genes
B) *myc* genes
C) *nif* genes
D) *nod* genes
Answer: D
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 23.3

36) The signaling factors of arbuscular mycorrhizae most likely gave rise to the signaling factors used in rhizobial nodule formation, yet less is known about arbuscular mycorrhizae because
A) they cannot be maintained in pure culture.
B) they are not as important for plant health.
C) no genetic systems have been developed for fungi.
D) they are not as important for plant health and no genetic systems have been developed for fungi.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.4

37) During the Jurassic Period, several different mammalian lineages independently evolved an herbivorous lifestyle for obtaining energy. This has resulted in
A) different digestive patterns in herbivorous animals, some that depend on gut microbiota to digest plant material and some that do not.
B) different digestive patterns in herbivorous animals that all depend on gut microbiota to digest plant material.
C) the evolution of foregut fermentation, as seen in ruminants, as the only digestive pattern that depends on fermentative gut microbiota.
D) horizontal gene transfer of genes for glycoside hydrolases and polysaccharide lyases from bacteria to mammals.
Answer: B
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 23.12

38) In mycorrhizal mutualisms between plants roots and fungi
A) the plant supplies carbohydrates to the fungus and the fungus supplies phosphorus and nitrogen to the plant.
B) the plant supplies water to the fungus and the fungus supplies essential amino acids to the plant.
C) the plant protects the fungus from predation and the fungus supplies carbohydrates to the plant.
D) the fungus infects the plant roots, stimulating plant growth through myc factors that act as growth hormones in the plant.
Answer: A
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 23.3

39) Secondary symbionts in insects often live
A) inside cells only.
B) in hemolymph only.
C) inside bacteriocytes only.
D) inside cells, in hemolymph, or inside bacteriocytes.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.5

40) The termite hindgut is
A) entirely oxic.
B) entirely anoxic.
C) variable in oxygen concentration, with distinct microbial niches maintained by patterns of microbial oxygen consumption.
D) entirely oxic or entirely anoxic depending upon the diet of the termite.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.7

41) ________ in insects are intracellular bacteria that are usually localized to specialized organs within their host.
A) *Symbiodinium*
B) Arbuscules
C) Epibionts
D) Endosymbionts
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.6

42) Lichens are a mutualistic association of a fungus and either an alga or a cyanobacterium in which
A) the fungus protects the photosynthetic partner from erosion.
B) the fungus slowly engulfs the photosynthetic partner.
C) the fungus helps collect sunlight for the photosynthetic partner.
D) the phototroph provides phosphorus to the fungus.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.1

43) Aphids that feed on carbohydrate-rich but nutrient-poor foods obtain ________ from their endosymbiotic bacterial partners.
A) ATP
B) amino acids
C) ammonium
D) volatile fatty acids
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.6

44) *Aliivibrio fischeri* symbionts benefit the Hawaiian bobtail squid by
A) mimicking the light of the moon, which helps the squid avoid nocturnal predators.
B) providing essential amino acids missing from the squid′s diet.
C) degrading the cellulosic cell walls of the phytoplankton the squid eats.
D) providing essential amino acids missing from the squid′s diet and degrading the cellulosic cell walls of the phytoplankton the squid eats.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.8

45) The structural and trophic foundation of coral reefs is a mutualistic relationship between
A) cyanobacteria and sponges (Porifera).
B) chemolithotrophic bacteria and stony corals (Cnidaria).
C) dinoflagellates and stony corals (Cnidaria).
D) cyanobacteria and stony corals (Cnidaria).
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.11

46) Bacterial symbionts most often provide marine invertebrates in hydrothermal vents with
A) fixed nitrogen in the form of ammonia.
B) sulfate and other sulfur compounds.
C) cellulolytic enzymes to help digest plant material.
D) fixed carbon dioxide in the form of organic compounds.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.9

47) *Riftia* tube worms contain unusual hemoglobins that bind(s)
A) H_2S and O_2 in order to transport these chemicals to their bacterial symbionts.
B) high concentrations of O_2 produced by their photosynthetic symbionts.
C) sulfate to transport this ion away from their bacterial symbionts.
D) ammonia that their bacterial symbionts produce.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.9

48) Which of the following is TRUE of *Riftia* tube worm symbionts?
A) they have never been maintained in laboratory culture
B) they are obligate symbionts
C) they have a free-living stage
D) they have an unusually small genome, even for a symbiont
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.9

23.2 True/False Questions

1) Consortia of a phototrophic green sulfur bacteria and motile heterotrophs are found worldwide in freshwater stratified sulfidic lakes.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.2

2) Leguminous plants such as alfalfa, beans, clover, peas, and soybeans all benefit from symbiotic bacteria that leach vitamins into their roots.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.3

3) Bacteria associated with plants fix very small amounts of nitrogen, therefore they do NOT contribute significantly to the global N cycle.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.3

4) Several rhizobial species within a cross-inoculation group should be able to grow with particular legume species such as alfalfa. However, the same cross-inoculation group would be unable to associate with another legume species such as peas.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.3

5) Bacteroids are rarely found within plant cells.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.3

6) While rhizobia usually associate with plant roots, nodules can also be formed along the stems of leguminous plants.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.3

7) An individual tree can maintain a symbiotic relationship with many different species of mycorrhizae simultaneously.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.5

8) The mycelia of mycorrhizae can form underground nutrient networks where several trees are connected.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.5

9) Plant diversity is generally lower in environments where mycorrhizae are associated with plants due to strong coevolution, which leads to competitive exclusion.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.5

10) Bacterial cells themselves serve as major sources of protein and vitamins in hindgut fermenting animals such as rabbits and horses.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.12

11) *Archaea* are commonly present in rumens.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.13

12) Fungi and protists serve an important role in the anaerobic digestion of cellulose in ruminants.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.13

13) *Fibrobacteres* comprise the greatest proportion of the bacterial species in ruminant guts.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.13

14) Ruminants are foregut fermenters.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.13

15) Hindgut fermenters use a cecum to contain fermentative organisms.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.12

16) The reproductive systems of many insects can be manipulated by parasitic symbionts, which are passed down to each generation and can skew the sex ratio of their progeny.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.6

17) Both pathogenic bacteria and primary symbionts have a tendency to lose genes for catabolic pathways.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.6

18) Horizontal gene transfer from symbionts to the DNA of their host nematodes is rare.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.6

19) Most termites have intracellular symbionts.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.7

20) Gut symbionts are lost after each molting a termite undergoes.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.7

21) Due to the close interactions with their hosts, obligate symbionts generally have larger genome sizes compared to other non-symbiotic bacteria.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.9

22) Microbial symbionts that are horizontally transferred show greater genome reduction than those that are heritable.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.6

23) Lichens always consist of a cyanobacterium and a fungus.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.1

24) *Photorhabdus* is a bacterial symbiont of insect nematodes.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.10

25) Coral bleaching is primarily caused by antimicrobials that disrupt the mutualistic relationships formed between corals and their bacterial symbionts.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.10

23.3   Essay Questions

1) Both lichens and corals are symbioses involving phototrophic microorganisms. Compare and contrast lichens and corals, including the types of organisms involved, the role of each partner, the specificity of the interaction, and the method of transmission for each symbiosis.
Answer: Lichens are found on house roofs, tree trunks, rocks, and soils. A lichen contains a fungus, which attaches to the surface and provides its mutualistic partner, either an alga or cyanobacterium, with increased access to inorganic nutrients. The fungus also prevents desiccation of the phototrophs by sequestering water and protects the phototroph from erosion in exposed environments. The phototroph produces organic carbon (sugars), which benefit the fungus. Some cyanobacteria also provide a fixed nitrogen source to the fungi. Many different fungi form associations with only a few different phototrophs, and the same phototroph can form an association with multiple fungi.
Corals are similar to lichens in that the cnidarian coral body provides protection for the phototroph (a dinoflagellate). The two partners also exchange the same nutrients – organic carbon in exchange for inorganic nutrients (N, P, and CO_2). However, the coral provides an additional benefit to the phototroph by increasing the collection of solar radiation for photosynthesis. Coral symbioses are also different in that the association between the coral host and the dinoflagellate symbiont is much more specific. Transmission of the symbiont is vertical – the symbiont is present in the egg before it is released from the parent, although free-living forms of the symbiont do survive outside of the host. The partners have specific recognition signals and specialized structures to coordinate growth and nutritional exchange.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 23.10

2) Genome sequencing of the bacterial symbionts of animals has allowed us to compare the genomic features of several symbionts. The following data was collected on two insect symbionts:

Symbiont A: 0.4 Mb genome, 20% G+C content, missing or incomplete peptidoglycan synthesis and nucleotide synthesis pathways.
Symbiont B: 1.8 Mb genome, 40% G+C content, peptidoglycan and most nucleotide synthesis pathways complete.

What can you deduce about these two insect symbionts based on their genomes?
Answer: Symbiont A is likely a vertically transferred symbiont because its genome is much smaller, has low G+C content, and is missing many biosynthetic genes. The stable transmission of a vertically transferred symbiont means that they no longer have a free-living form. Therefore, over time, functions that can be supplied by the host or are no longer necessary are lost. The adenine plus thyme content increases because common mutations that convert G-C pairs to A-T pairs are not fixed in symbionts that have fewer DNA repair enzymes to fix these mutations. Symbiont B is most likely a more recent association that is horizontally transferred between hosts instead of vertically transferred from parent to offspring. It may also still have a free-living form because most biosynthetic pathways are still complete.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 23.6

3) Based on 16S RNA phylogenetic data, what phyla are important in the ruminant gut? Give an example of how diet affects ruminant health.
Answer: The phyla that are most heavily represented by species in the ruminant gut are Firmicutes, Bacteroidetes, and Euryarchaetoa. Changes in the microbiota can have significant effects on health. One example is that overgrowth of *Streptococcus bovis*, which can occur when the diet is changed to include much more grain, can lead to acidosis. *S. bovis* produces lactic acid, which lowers the rumen pH, preventing normal digestion and potentially leading to serious complications or even death. As another example, inoculation of *Synergistes jonesii* can prevent animals from being poisoned by mimosine by-products in *Leucaena*.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 23.13

4) The following quote is taken from a university extension publication for farmers:

"Grain poisoning, also known as grain overload or lactic acidosis, is usually the result of stock consuming large quantities of grain or pellets to which they are unaccustomed. Pasture-fed cows or feedlot cattle not yet adapted to grain may become acutely ill or die after eating only moderate amounts of grain, whereas stock accustomed to diets high in grain content may consume large amounts of grain with little or no effect."

What is happening in the digestive system of the livestock that causes grain poisoning and why does a gradual shift in diet avoid this problem?
Answer: A major shift in diet, where the rumen flora must suddenly ferment a different substrate, results in different products formed. When ruminant animals suddenly consume grain instead of grass, the starch in the grain causes *Streptococcus bovis* to grow and produce large quantities of lactic acid. Lactic acid is a stronger acid than other volatile fatty acids that are usually produced (e.g., butyric and proprionic acid). When the pH of the rumen decreases, blood acidosis can ensue (which can be lethal). If the ruminant switches slowly to a grain diet, the fermentative bacteria in the rumen adapt and bacteria that produce the normal fatty acids are selected for allowing rumen functions to continue normally.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 23.13

5) What is the difference between a primary and secondary insect symbiont? Propose how you might use an insect symbiont to control the spread of an insect-borne disease.
Answer: Primary symbionts are microorganisms that are required for the host insect to replicate. Secondary symbionts are not required for the insect to replicate, so they are not found in every member of the insect population. There are many different ways that an insect symbiont could be manipulated to control the spread of an insect-borne disease. One way is to use a secondary symbiont that manipulates the reproduction of the insect host, such as *Buchnera* species that sterilize uninfected females after mating with an infected male. Another possibility would be to use an antimicrobial to inhibit the primary symbiont of an insect such that the death of the primary symbiont would stop the insects from reproducing.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 23.6

6) How do Hawaiian bobtail squids prevent all bacteria EXCEPT *Allivibrio fischeri* from colonizing their light organ?
Answer: Firstly, the mucus produced in the light organ specifically aggregates gram-negative bacteria (*Allivibrio fischeri* is gram-negative). In addition, the squid excrete toxic levels of nitric oxide (NO) around the organ. *A. fischeri* contains NO-inactivating enzymes that allow it to tolerate NO exposure and colonize the organ. NO is continually produced in the light organ, thus preventing other bacterial species from colonizing the tissue.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.8

7) Describe the mutualistic relationship between chemolithotrophic bacteria and tube worms in hydrothermal vents.
Answer: The tube worm has a special form of hemoglobin that binds to both hydrogen sulfide and oxygen. The symbiotic bacteria in the trophosome of the tube worm oxidizes the hydrogen sulfide with the oxygen to generate energy. The energy from sulfide oxidation is then used to fix carbon dioxide, and the organic carbon is secreted to provide a food source for the tube worm. In addition, the tube worm has high CO_2 content in its blood, which serves as a carbon source for the bacteria to fix. This allows the large tube worms to grow on the primary production of the sulfur-oxidizing bacteria with no dependence on sunlight or photosynthesis for primary production.
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.9

8) Termites lose gut symbionts when they molt. How are they able to maintain a stable gut microbiota?
Answer: Termites are very social organisms. When they molt, they can probably obtain new gut symbionts through close contact with other termites.
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.7

9) What is coral bleaching and how might corals be able to adapt to avoid the destruction of coral reefs?
Answer: Coral bleaching occurs when the phototrophic symbiont in the coral dies, revealing the underlying colorless (white) limestone skeleton of the coral. It is thought that the main factors in coral bleaching are an increase in both water temperature and high UV irradiance. Global climate change is increasing the sea surface temperature. The phototrophs are very sensitive to changes in temperature and UV irradiance, which may cause an increase in reactive oxygen species that causes the coral to rapidly kill and expel the unhealthy photobionts. Corals and their symbionts, may, however be able to adapt to increased temperatures. There are over 150 different phylotypes of the dinoflagellate symbiont *Symbiodinium*, many of which may have different heat and UV stress tolerance. Corals are capable of either switching symbionts through horizontal transmission or swapping symbionts by allowing a rare symbiont to take over the population. Understanding which symbionts are capable of colonizing hosts that have lost their initial symbionts can have predictive value for the future of reef ecosystems.
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 23.11

10) The attine ant is entirely dependent on multiple microbial species for its source of food. What microbial species are involved and what is their role in this multi-partner symbiosis?
Answer: Attine ants purposefully cultivate fungi gardens, where a particular species of fungus degrades small leaves and provides a food source for the ants. However, there is a parasitic fungus present that inhibits the garden fungus. An *Actinobacteria* species forms a second mutualistic relationship with the attine ant. The *Actinobacteria* grows on the surface of the ant and secretes an antifungal compound that inhibits that parasitic fungus, thus protecting the garden fungus from the parasite. Yet another microbial partner, a black yeast, also grows on the ant's body and inhibits the actinobacterium′s antifungal production.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 23.6

11) How are symbiotic relationships important in agriculture and food production? Use at least two different examples to support your answer and explain how understanding these symbioses could help produce more food.
Answer: Answers can use any two of the following symbioses: i) Root nodules (22.3); ii) arbuscular mycorrhizae (22.5); iii) *Azolla-Anabaena* (22.3); iv) *Agrobacterium* (22.4); or v) ruminant animals (22.7). The answers should explain the symbiosis in sufficient detail such that a conclusion can be reached about how the symbiosis increases production of food. For example, root nodules result in increased N supply to legumes (bean crops and alfalfa) without fertilizer; arbuscular mycorrhizae increase N and P supply to plants such as wheat and oats; *Anabaena* increases N supply to rice without fertilizer; *Agrobacterium* can be used to engineer crops in many ways to increase yield, disease resistance, or drought tolerance; and understanding ruminant animal biology could improve the health of the ruminant animals and increase production of milk and/or meat.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 23.4